UE24CS242B - OS Orange Level 2

PES2UG24CS605 - Yadunandana Reddy M - 4J

Repository- https://github.com/Spinicle/PES2UG24CS605-PES-VCS

Phase 1

```
osboxes@tetsuya:~/PES/OS/PES2UG24CS605-PES-VCS$ make test_objects
./test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
osboxes@tetsuya:~/PES/OS/PES2UG24CS605-PES-VCS$
```

```
osboxes@tetsuya:~/PES/OS/PES2UG24CS605-PES-VCS$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
osboxes@tetsuya:~/PES/OS/PES2UG24CS605-PES-VCS$
```

Phase 2

```
osboxes@tetsuya:~/PES/OS/PES2UG24CS605-PES-VCS$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
osboxes@tetsuya:~/PES/OS/PES2UG24CS605-PES-VCS$
```

```
osboxes@tetsuya:~/PES/OS/PES2UG24CS605-PES-VCS$ xxd .pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
00000000: 626c 6f62 2036 0068 656c 6c6f 0a         blob 6.hello.
```

Phase 3

```
osboxes@tetsuya:~/PES/OS/PES2UG24CS605-PES-VCS$ ./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
  untracked:  test_tree
  untracked:  file.txt
  untracked:  tree.h
  untracked:  test_sequence.sh
  untracked:  index.c
  untracked:  bye.txt
  untracked:  commit.h

osboxes@tetsuya:~/PES/OS/PES2UG24CS605-PES-VCS$
```

```
osboxes@tetsuya:~/PES/OS/PES2UG24CS605-PES-VCS$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776450359 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776450359 6 file2.txt
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776450244 6 temp.txt
osboxes@tetsuya:~/PES/OS/PES2UG24CS605-PES-VCS$
```

Phase 4

```
osboxes@tetsuya:~/PES/OS/PES2UG24CS605-PES-VCS$ ./pes init
echo "Hello" > hello.txt
echo "Hello" > hello.txt
./pes add hello.txtitial commit"
./pes commit -m "Initial commit"
echo "World" >> hello.txt
echo "World" >> hello.txt
./pes add hello.txtd world"
./pes commit -m "Add world"
echo "Bye" > bye.txt
echo "Bye" > bye.txt
./pes add bye.txtAdd farewell"
./pes commit -m "Add farewell"
./pes log
./pes log
Initialized empty PES repository in .pes/
Committed: 996672a31eae... Initial commit
Committed: 824830f74273... Add world
Committed: aed5828b080d... Add farewell
commit aed5828b080d23e602a6b1f4ba6a394b7392c5cf5e9cd6787662eedc2943169d
Author: PES User <pes@localhost>
Date:   1776450407

    Add farewell

commit 824830f74273811aa465a8f8ccfce1323888dc4adae464d5c20fd1840e96127e
Author: PES User <pes@localhost>
Date:   1776450407

    Add worldell

commit 996672a31eae791786364dac9c0f5fae329916a773aed9978ec84f5ef8beef89
Author: PES User <pes@localhost>
Date:   1776450407

    Initial committial commiQ

osboxes@tetsuya:~/PES/OS/PES2UG24CS605-PES-VCS$
```

```
osboxes@tetsuya:~/PES/OS/PES2UG24CS605-PES-VCS$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/20/0d09d78508ca1f21649eab5001448ccd9eed88ca79ded8310a9c71a0c0ba58
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/48/1407f7237f7bd749dca31e2d8422bafe211741d5556f007e0faf821ffc7179
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/76/7689aa3673cfd1764725a5df1d48b0e252dfb9c1ae3ee566cf96a6ea62fd82
.pes/objects/82/4830f74273811aa465a8f8ccfce1323888dc4adae464d5c20fd1840e96127e
.pes/objects/99/6672a31eae791786364dac9c0f5fae329916a773aed9978ec84f5ef8beef89
.pes/objects/ae/d5828b080d23e602a6b1f4ba6a394b7392c5cf5e9cd6787662eedc2943169d
.pes/objects/bb/3e28f19aac58d0b589b8e871604baa97ca88d1dbed93896536dc4a5b93687b
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/refs/heads/main
osboxes@tetsuya:~/PES/OS/PES2UG24CS605-PES-VCS$
```

```
osboxes@tetsuya:~/PES/OS/PES2UG24CS605-PES-VCS$ cat .pes/refs/heads/main
cat .pes/HEAD
aed5828b080d23e602a6b1f4ba6a394b7392c5cf5e9cd6787662eedc2943169d
ref: refs/heads/main
osboxes@tetsuya:~/PES/OS/PES2UG24CS605-PES-VCS$
```

Test-Integration

```
osboxes@tetsuya:~/PES/OS/PES2UG24CS605-PES-VCS$ make test-integration
commit edc2483f594687ec64b179bcb5c6daa21ccc29379440dd6408957d5eb081102d
Author: PES User <pes@localhost>
Date:   1776450492

    Add farewell

commit 52685bdecd84468694540b089972fe92cce6000b77be75b49c6a03f7c0e5dade
Author: PES User <pes@localhost>
Date:   1776450492

    Update file.txt

commit a6c181f6813223555c33dbfdbc8f54f7405834d43b17f5c5411e0f5d7f1938b8
Author: PES User <pes@localhost>
Date:   1776450492

    Initial committial commiQ


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
edc2483f594687ec64b179bcb5c6daa21ccc29379440dd6408957d5eb081102d

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/52/685bdecd84468694540b089972fe92cce6000b77be75b49c6a03f7c0e5dade
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/a6/c181f6813223555c33dbfdbc8f54f7405834d43b17f5c5411e0f5d7f1938b8
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/ed/c2483f594687ec64b179bcb5c6daa21ccc29379440dd6408957d5eb081102d

=== All integration tests completed ===
osboxes@tetsuya:~/PES/OS/PES2UG24CS605-PES-VCS$
```

Phase 5

Q5.1

A branch is just a file that stores a commit hash. To implement pes checkout <branch>, the system must first update the HEAD file to point to the selected branch. Then it should read the commit hash stored in that branch file and retrieve the corresponding tree.

After that, the working directory must be updated to match the snapshot represented by that tree. This involves removing files that are not present in the target commit and restoring files that are present in it.
The complexity arises from safely modifying the working directory. Files may already exist, may be modified, or may conflict with the target branch. Handling these cases without losing user data makes the operation difficult.

Q5.2
To detect a dirty working directory, the system compares the current state of files with the index.
For each tracked file, the stored hash in the index is compared with the actual file contents or metadata. If they differ, the file has been modified but not staged.
Then, the system compares this with the target branch. If the same file differs in the target commit, switching branches would overwrite local changes. In such cases, checkout must be prevented to avoid data loss.

Q5.3
Detached HEAD means that HEAD contains a commit hash directly instead of pointing to a branch.
If a commit is made in this state, the new commit is created but is not referenced by any branch. As a result, it becomes unreachable when the user switches away.
However, the commit still exists in the object store. The user can recover it if they know the hash by creating a new branch that points to that commit.

Phase 6

Q6.1
To remove unreachable objects, the system first identifies all reachable objects starting from branch heads.
It traverses each commit, marking the commit itself, its associated tree, and recursively all blobs and subtrees. A set or hash table can be used to efficiently track visited objects.
After this traversal, any object not present in the set is considered unreachable and can be safely deleted.
In a repository with 100000 commits and 50 branches, many commits will overlap across branches, so the number of unique objects visited will be significantly less than the total possible number.

Q6.2
Running garbage collection during a commit is risky due to race conditions.
A commit operation may create new objects but not yet update references. If garbage collection runs at this moment, it may treat those new objects as unreachable and delete them.
This can lead to repository corruption because the commit may later reference objects that no longer exist.
Git avoids this by using locking mechanisms and ensuring garbage collection runs only when the repository is in a stable state.